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Integrating Digital Twins, Data Management and Artificial Intelligence for Predictive Maintenance in Aviation: A Comprehensive Review

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ABSTRACT

Predictive maintenance (PdM) plays a critical role in enhancing safety, operational efficiency and cost-effectiveness in the aviation industry by enabling condition-based maintenance strategies instead of traditional schedule-driven approaches. This paper presents a systematic scoping review of the core technologies underpinning data-driven PdM in aviation, with a particular focus on digital twin (DT) systems, engineering data management and artificial intelligence (AI) algorithms, and their integration across the PdM pipeline. The reviewed studies are systematically categorised according to three primary data types used in aviation PdM—time-series sensor data, graphical data and natural language data—together with their associated methods for data collection, preprocessing, storage and analysis. In addition, the review analyses AI-based approaches for remaining useful life estimation and fault detection, highlighting commonly adopted models and benchmark datasets such as C-MAPSS. Key challenges identified in the literature include data heterogeneity, real-time processing constraints, scalability and cybersecurity risks. Emerging solutions, including multimodel database architectures and fog–cloud hybrid computing frameworks, are discussed as enablers of robust and scalable PdM systems. By providing an integrated and aviation-specific perspective, this review offers researchers and practitioners a structured foundation for the design, development and deployment of DT–enabled PdM systems in safety-critical aviation environments.

1 | Introduction

In the aviation industry, repair and maintenance activities are critical to ensuring operational safety and efficiency. These activities encompass routine inspections of aircraft components, fault detection and repair and preventive measures aimed at avoiding potential damage or unplanned operational downtime [1]. Traditionally, such practices have been based on scheduled inspections and component replacements performed at fixed intervals, irrespective of the actual condition of the equipment. Although this approach provides systematic oversight, it often leads to unnecessary maintenance actions and increased operational costs.

Predictive maintenance (PdM) provides a data-driven framework that utilises real-time aircraft system data and predictive analytics to anticipate potential failures and support timely condition-based maintenance. By identifying faults prior to their occurrence, PdM enhances operational safety, reduces the risk of in-flight failures, lowers maintenance costs and aircraft downtime and mitigates unplanned service disruptions, thereby improving overall operational reliability [2–4].

A recent study [3] reports that approximately 49% of aviation accidents are attributed to pilot error, 23% to mechanical failure and the remaining 28% to factors such as adverse weather, sabotage, bird strikes, mid-air collisions, aircraft overloading and ground crew errors. In this context, PdM plays a critical role in enhancing safety and operational performance by integrating condition-based and fleet-wide monitoring techniques. It proactively addresses potential issues, including engine failures, structural degradation, fuel shortages and navigational challenges, thereby reducing the likelihood of unscheduled downtime and improving overall operational reliability.

Digital twin (DT) technology, a core component of Industry 4.0, complements PdM by enabling the creation of high-fidelity virtual replicas of physical aircraft systems that support real-time modelling, monitoring and data-driven decision-making. A DT integrates the physical system, its virtual counterpart and bidirectional data exchange by leveraging IoT sensor streams, historical data and physics-based simulations to continuously assess the health of critical components such as engines, wings and landing gear [5]. This capability facilitates proactive fault detection, early anomaly identification and accurate estimation of remaining useful life (RUL), thereby enhancing maintenance planning, aircraft safety and operational performance. Furthermore, by simulating real-world operational scenarios, DTs support lifecycle management, innovative aircraft design and performance optimisation, underscoring the transformative role of data-centric DT–PdM integration in aviation, where data quality and effective utilisation critically determine overall system effectiveness [2, 6, 7].

Modern aircraft generate vast volumes of heterogeneous operational data, providing critical insights into aircraft usage, management and maintenance requirements. For example, the Airbus A380 is equipped with up to 25,000 sensors that generate terabytes of data per flight; these data are further complemented by graphical information derived from images and videos as well as unstructured text from maintenance logs and pilot reports [2]. The volume, velocity and variety of such datasets firmly place aviation data within the big data paradigm, posing significant challenges for real-time analysis whilst simultaneously offering a rich foundation for predictive maintenance. When effectively utilised, these data enable proactive fault diagnosis, accurate estimation of RUL and early anomaly detection, ultimately enhancing maintenance efficiency, operational reliability and aviation safety [8, 9]. Although the increasing availability of heterogeneous aviation data has created new opportunities for PdM, their effective exploitation requires the coordinated integration of engineering data management, DT technologies and artificial intelligence (AI). Engineering data management ensures data quality, consistency and traceability across the asset lifecycle, DTs enable continuous synchronisation between physical systems and their virtual representations, and AI algorithms leverage this integrated environment to support fault diagnosis, RUL estimation and anomaly detection. In the absence of such integration, PdM solutions tend to remain fragmented and face significant challenges in scalability and practical deployment within safety-critical aviation contexts [10].

Existing studies address various aspects of AI-based PdM but often do not comprehensively consider the integration of DT technologies or engineering data management, particularly within the aviation domain. One study [11] surveys deep learning (DL) based PdM strategies by categorising methods, reviewing algorithms and discussing datasets and hardware; however, its focus spans multiple industries rather than being specific to aviation. Another study [12] investigates the application of generative AI (GAI) for PdM in aviation, addressing the scarcity of run-to-failure data through synthetic data generation using generative adversarial networks (GANs) and variational autoencoders (VAEs) to improve predictive accuracy. Nevertheless, it does not examine other AI algorithms or DT-based applications. A third study [13] reviews AI certification for airborne systems, classifying applications according to criticality and methodological complexity and briefly highlighting issues related to data quality, representativeness and dependency; however, it does not explore DT concepts.

Several review papers have examined individual data modalities in aviation maintenance contexts, without providing a holistic view across multiple data sources. A systematic literature review [14] provides an in-depth analysis of vision-based approaches for aircraft inspection, with a particular emphasis on maintenance, repair and overhaul (MRO) activities. It evaluates 27 primary studies—including experimental investigations, conceptual frameworks and prior reviews—to assess the integration of computer vision and robotics in automating visual inspection processes. The review identifies key limitations, such as the lack of standardised requirements, limited solution validation and the absence of fully integrated intelligent inspection systems; however, its scope is confined to image- and video-based techniques and does not consider nonvisual methods unless combined with vision-based systems. Another review [15] examines the use of unmanned aircraft systems (UASs) for visual inspection of aircraft structures and airport infrastructure, analysing technical, operational and regulatory feasibility. It covers multirotor platforms, onboard sensors, such as high-resolution, thermal and light detection and ranging (LiDAR), image post-processing techniques (e.g., CNNs and GANs) and outputs, such as three-dimensional models, orthomosaics and thermal maps, highlighting real-world case studies in automated flight planning, damage detection and inspection reporting. Similar to the previous review, it does not address nonvisual techniques unless they are integrated with UAS- or image-based methods. A separate study [16] systematically examines natural language processing (NLP) applications in aviation safety, analysing 30 peer-reviewed studies published between 2010 and 2022 that focus on incident and accident report analysis and air traffic control communications. It evaluates deep learning, topic modelling and speech recognition approaches, identifying challenges, such as linguistic ambiguity, limited multilingual support, data scarcity and real-time processing constraints, whilst also noting emerging trends in transformer-based models such as bidirectional encoder representations from transformers (BERT) and large language models (LLMs) from the GPT-series. This review underscores the need for domain-specific datasets and further research on robust, real-time and multilingual NLP systems for aviation safety.

Motivated by the limitations of prior studies, which often focus on isolated aspects of data utilisation, AI or predictive maintenance without providing a comprehensive guide, an integrated perspective for aviation applications is presented. The role of DT technology and data utilisation in enabling PdM within the aviation domain is examined. It presents an integrated aviation-specific perspective by jointly considering three key areas—engineering data management, DT systems and AI algorithms—across the entire PdM pipeline. The discussion encompasses DT architectures, PdM strategies, data types, storage and database selection, preprocessing methods and AI-based data analysis techniques, thereby providing a comprehensive overview of the end-to-end PdM process. In contrast to most existing studies and surveys, which typically address only one or two aspects in isolation, such as focusing on AI techniques without accounting for data management challenges or examining digital twin technologies without considering practical implementation constraints, including storage and cloud-related issues—a unified framework is presented to provide an integrated perspective on aviation predictive maintenance systems for both academic researchers and industry practitioners.

The main contributions of this paper are summarised as follows:

- A comprehensive review of PdM in the aviation industry, with a particular emphasis on data-driven methodologies and their implications for safety, operational efficiency and cost reduction.
- A detailed examination of DT technology, including its functional architecture, integration with cloud and blockchain platforms and associated security considerations relevant to aviation applications.
- An analysis of three primary data types applicable to aviation PdM—time-series, graphical and natural language data—together with relevant preprocessing and cleaning techniques as well as suitable storage solutions.
- A systematic survey of AI algorithms tailored to each data type, highlighting methods for RUL estimation and fault detection, supported by comparative evaluations based on real-world datasets and performance metrics.
- A discussion of key challenges in engineering data management and AI deployment for PdM, along with potential solutions, including multimodel databases, fog–cloud hybrid computing infrastructures and privacy-enhancing technologies.

This paper is organised as follows. Section 2 describes the research methodology. Section 3 introduces DT technology, outlining its definition and industrial applications. Section 4 reviews PdM and the role of data, with particular emphasis on data storage approaches, including database systems, cloud computing and security considerations. Section 5 highlights the importance of data cleaning, especially for sensor-generated time-series data that require preprocessing prior to analysis. Section 6 examines AI algorithms relevant to such data. Section 7 discusses the key challenges in the field. Finally, Section 8 presents the conclusion.

2 | Review Methodology

2.1 | Review Design, Information Sources and Search Strategy

This study follows a *systematic scoping review* methodology to provide a comprehensive and structured overview of engineering data management, digital twin technologies and artificial intelligence algorithms for predictive maintenance in aviation. The review process was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines to ensure transparency, reproducibility and methodological rigour.

The remainder of this section details the information sources and search strategy, the eligibility criteria for study inclusion and the multistage selection process and PRISMA-based workflow used to synthesise the final collection of studies.

A comprehensive literature search was conducted across major scientific databases, including IEEE Xplore, Scopus, Web of Science, ScienceDirect, SpringerLink and ACM Digital Library, ensuring broad coverage of peer-reviewed journal articles and conference proceedings. The search strategy combined keywords related to predictive maintenance, aviation systems, digital twin technology and artificial intelligence (e.g., '*predictive maintenance*', '*aviation*', '*aircraft*', '*digital twin*', and '*deep learning*') using Boolean operators to refine the queries for English-language publications.

Studies were included if they focused on predictive maintenance or digital twin systems within aviation, presenting technical methods, architectures or empirical results. Conversely, studies were excluded if they were unrelated to aviation, focused solely on nontechnical aspects or lacked sufficient scientific rigour.

2.2 | Study Selection Process and PRISMA Workflow

The study selection process followed PRISMA-ScR recommendations across multiple stages. Initially, records were identified through database searches and duplicate entries were removed. Titles and abstracts were screened to exclude clearly irrelevant studies, followed by a full-text review to determine eligibility based on the predefined inclusion and exclusion criteria.

As illustrated in the PRISMA flow diagram (Figure 1), 1236 records were identified through database searches, with 48 additional records found via manual screening. After removing 312 duplicates, 972 records remained for screening, of which 781 were excluded. Following full-text assessment of 191 articles, 108 were excluded for failing to meet technical or domain-specific criteria. Ultimately, 83 studies were included in the final qualitative synthesis.

3 | Digital Twin Technology

The remainder of this section provides a formal definition of DTs, explores their functional layered architecture and reviews their diverse applications across various industrial sectors.

3.1 | Definition of Digital Twin

As a dynamic virtual representation, a DT mirrors a physical entity in real time and generates actionable insights for analysis and decision-making [17]. The DT enables bidirectional data communication between a physical system and its virtual counterpart. It integrates real-time data with AI and data analytics to enhance performance, deliver functionalities and provide services [6, 18]. By leveraging historical data, machine learning models and simulation tools, DTs can predict potential failures before they occur. For instance, patterns in sensor data may reveal early signs of wear or impending component malfunction, facilitating timely predictive maintenance interventions. Figure 2 presents a high-level overview of a DT framework that supports PdM in aviation [10].

The effectiveness of a DT depends on the quality and volume of the data it receives, underscoring the importance of real-time integration. This integration is facilitated through IoT sensors and data processing systems that continuously capture, analyse and transmit information between physical and digital assets,

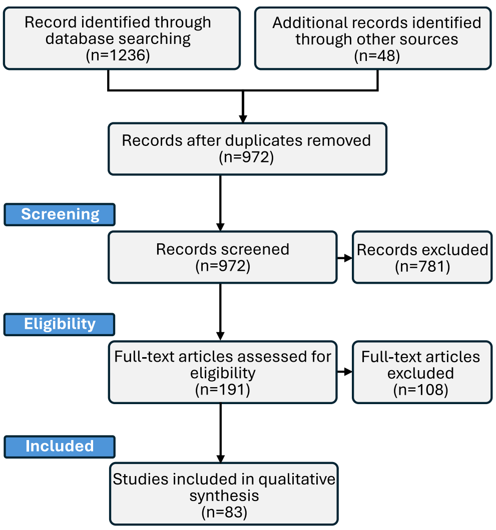

FIGURE 1 | PRISMA-based flow diagram illustrating the identification, screening, eligibility assessment and inclusion of studies in this review.

ensuring that the DT accurately reflects the current state of its physical counterpart [5].

A DT system generally comprises four functional layers, each serving a distinct role: data dissemination and acquisition, data management and synchronisation, data modelling and additional services and data visualisation and accessibility [19–21]. The first layer is responsible for collecting data from the physical environment (e.g., via sensors) and generating control commands for the physical system. It forms the foundation of the DT system by providing essential data to the upper layers. The second layer, data management and synchronisation, ensures data consistency and performs preprocessing before transmitting information to higher-level layers, thereby facilitating continuous interaction between the physical and digital environments. The third layer, data modelling and additional services, defines the states, behaviours and geometric structures of assets through digital models, supporting detailed modelling, simulation, analysis and performance prediction of physical assets. The final layer, data visualisation and accessibility, provides end-users with a visual representation of the physical system, enabling informed decision-making through simulation outputs and analytical insights. Collectively, these layers enable the seamless integration of physical and digital components, forming the core of an effective DT system. This layered architecture allows DTs to be tailored to specific industrial requirements by emphasising relevant functionalities.

In the following section, we examine the implementation of these capabilities across diverse industries, which collectively enhance operational efficiency, safety and innovation.

3.2 | Applications of Digital Twins in Various Industries

The advent of Industry 4.0, together with advances in big data analytics, machine learning and deep learning paradigms, has accelerated the adoption of DT technology across industrial domains [22, 23]. A primary objective of DTs is to provide a comprehensive understanding of the behaviour and condition of physical assets. By continuously collecting extensive real-time data, DTs offer near-instantaneous representations of asset status, enabling engineers and operators to monitor performance, detect deviations and make informed decisions. This capability is particularly critical in high-risk sectors, such as aviation, where real-time insights support proactive failure prevention [24].

The primary motivation for developing DTs is to enhance product lifecycle management in manufacturing industries [25].

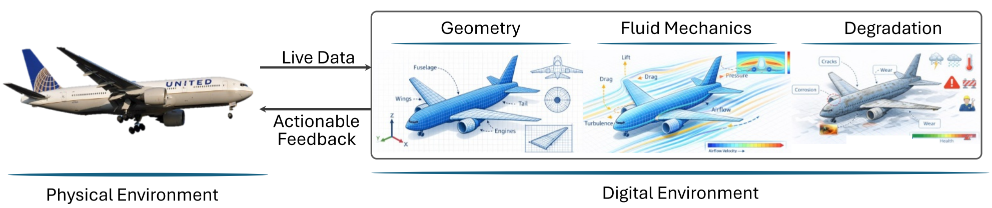

FIGURE 2 | High-level overview of using digital twin technology supporting aviation-related predictive maintenance.

The implementation of DTs facilitates real-time equipment monitoring and PdM, thereby minimising unplanned downtime and extending machinery lifespan. By simulating production scenarios and leveraging high-quality data, they enable real-time process adjustments, improve efficiency, reduce waste and support early defect detection with prompt corrective actions [26].

In healthcare, DTs represent a transformative approach by leveraging smart wearable devices and biosensors to construct high-fidelity digital replicas of the human body. These digital models enable advanced health monitoring, precision surgery, and telemedicine, thereby transforming patient care. The use of DTs enables the simulation of patient organs through real-time health data, facilitating personalised treatment plans and predictive medical interventions. Furthermore, they support the development of interconnected healthcare systems, improving patient data tracking and management across diverse medical services [27, 28].

The integration of DT technology is a key component in the development of smart cities, providing applications that enhance urban management and improve citizens' quality of life. By generating accurate virtual models of urban environments, DTs enable city planners to simulate scenarios, such as changes in land use, population density and environmental impact, thereby supporting data-driven decision-making for sustainable development [29]. Real-time monitoring and predictive maintenance of critical infrastructure are further enhanced by DTs, enabling administrators to proactively address potential issues in bridges, roads, and municipal services. Additionally, by simulating traffic patterns and congestion points, DTs support the optimisation of public transportation and traffic management, enhancing traffic flow and reducing commute times [30].

In summary, DT technology is transforming various sectors by enhancing operational efficiency, predictive capabilities and overall system intelligence. Industries, such as manufacturing, aerospace, automotive, healthcare, energy, transportation, construction and agriculture, are increasingly adopting DTs to drive innovation and optimise processes. As DT applications continue to evolve, they are expected to assume an even more prominent role in advancing industrial and operational excellence. Table 1 provides an overview of DT applications across selected industries.

According to a study [6], the significance of DT technology in PdM lies in its capacity to leverage real-time data and high-fidelity models for enhanced monitoring, simulation and predictive analytics, thereby enabling accurate analysis of asset behaviour and improved decision-making in maintenance operations. In the following section, we examine PdM in the aviation industry and its integration with DTs, utilising diverse data types, including time-series sensor data, graphical data and natural language data. Time-series data are collected from engine metrics, vibration, pressure and other telemetry, whereas graphical data are obtained from fuselage inspections and structural assessments. Natural language data comprise pilot logs, maintenance reports and other textual records. Collectively, these heterogeneous sources support the creation of comprehensive and intelligent digital representations of physical assets, which are central to effective predictive maintenance.

4 | Aircraft Predictive Maintenance

This section subsequently examines the primary data types used in aviation PdM, discusses suitable database management systems, evaluates cloud-based infrastructures and addresses critical security concerns and solutions.

4.1 | Overview of Predictive Maintenance in Aviation

PdM is transforming the aviation industry by leveraging advanced data analytics, IoT technologies and innovative methodologies to enhance aircraft reliability, operational efficiency and availability while minimising unnecessary maintenance actions and associated costs [2]. At its core, PdM integrates condition-based maintenance with predictive models, with data collection playing a fundamental role.

TABLE 1 | Summary of digital twin applications across industries.

Ref.	Industry	Applications	Benefits
[27, 31]	Healthcare	Personalised treatment planning; virtual organ/patient models for diagnostics and training	Improved outcomes; enhanced surgical precision and reduced trial-and-error
[32]	Automotive	Vehicle design and testing; real-time monitoring for autonomous driving	Faster innovation; improved safety and enhanced predictive maintenance
[33]	Energy	Grid monitoring and optimisation; maintenance of wind turbines and power plants	Reduced energy waste; higher reliability and lower operational risk
[29]	Smart cities	Traffic optimisation; waste management and energy-consumption monitoring	Reduced congestion; improved urban planning and increased sustainability
[34, 35]	Agriculture	Crop-growth monitoring; precision farming for resource optimisation	Higher yield; optimised use of water/fertilisers; reduced waste
[36]	Aerospace	Aircraft performance monitoring; predictive maintenance	Higher reliability; lower operating cost and improved flight safety

The primary objective of PdM in aviation is to enhance safety and reliability through advanced diagnostic and prognostic capabilities, with DT technology serving as a key enabler. The use of DTs generates virtual replicas of aircraft systems that continuously simulate and monitor real-time operational conditions by analysing data from onboard sensors and external sources [37]. Leveraging machine learning models, statistical analyses and physics-based simulations, DTs evaluate the health status of critical components, such as engines, wings and landing gear, enabling early failure prediction and accurate estimation of RUL. This capability not only reduces maintenance costs and operational downtime but also supports proactive maintenance scheduling in compliance with stringent aviation safety regulations, including those of the European Union Aviation Safety Agency (EASA) and the Federal Aviation Administration (FAA) [38].

Figure 3 illustrates the general functional model of PdM, in which data acquisition and processing constitute the initial stage. Accordingly, identifying and categorising available data types is a fundamental prerequisite for initiating effective PdM operations.

4.2 | Different Data Types

Data play a pivotal role in transforming aviation PdM by altering how aircraft systems are monitored and maintained, whilst simultaneously enhancing operational efficiency and safety [39]. The widespread deployment of onboard sensors allows the collection of large volumes of operational data, including vibration, temperature and pressure signals, which are analysed to identify degradation patterns and trends. For instance, a study [7] demonstrates the effectiveness of vibration-based machine learning models for fault diagnosis and RUL prediction in multirotor unmanned aerial systems (UAS), underscoring the value of data-driven maintenance strategies.

The transition from traditional preventive maintenance to PdM represents a big-data-driven paradigm shift in aviation as real-time data enable maintenance actions tailored to actual operating conditions rather than fixed schedules, which often result in unnecessary replacements or unforeseen failures [2, 6]. However, this shift introduces significant data management challenges due to the massive volume of information generated by modern aircraft, thereby highlighting the need for advanced analytics platforms and sophisticated data science techniques to extract meaningful insights. For example, a twin-engine Boeing 737 flying an average of 6 h generates approximately 20 terabytes of data per engine per hour [40]. Figure 4 illustrates this scale over the course of a year.

Generally, three primary data types are used in aviation PdM [2]:

- Time-series data.
- Graphical data.
- Natural language data.

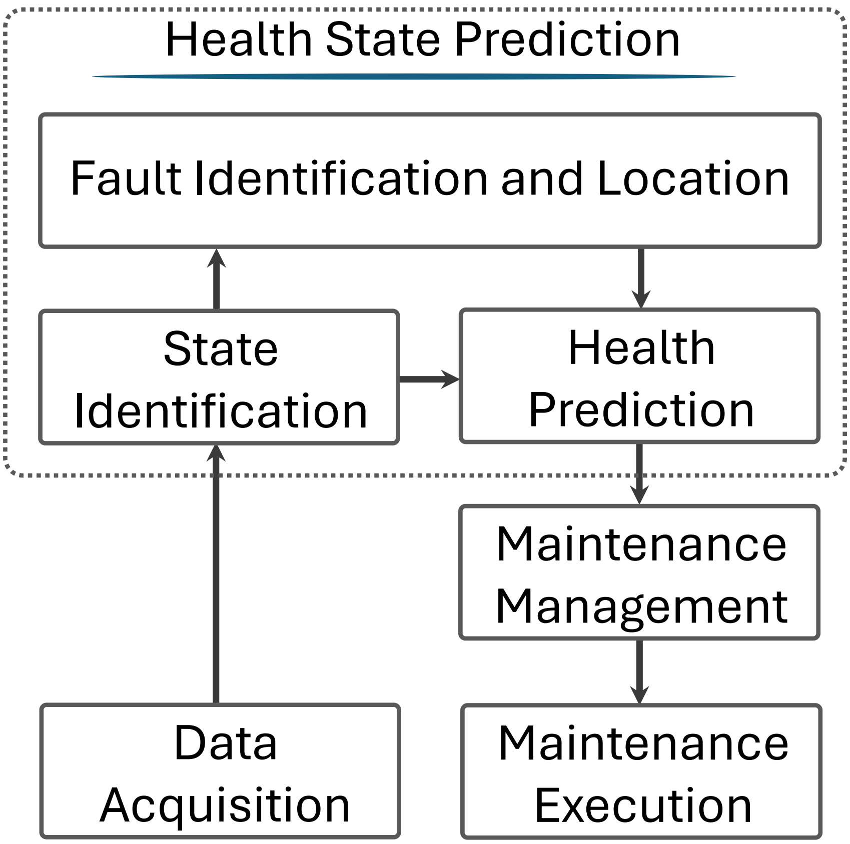

FIGURE 3 | The system function model of predictive maintenance.

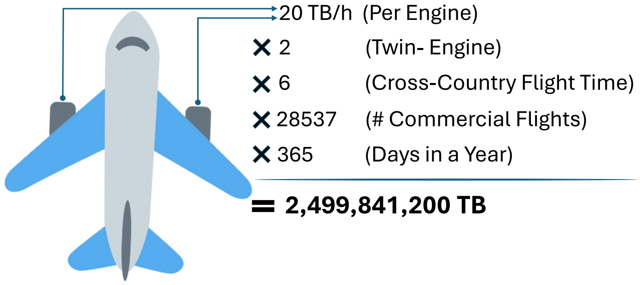

FIGURE 4 | The volume of data collected from the two engines sensors during flights conducted in 1 year with an average of 6 h of flight time per day [40].

Time-series data consist of sequential observations recorded at regular intervals, such as temperature, pressure and vibration measurements, collected from aircraft sensors, which are analysed to monitor system performance trends and detect anomalies. Graphical data encompass images and videos acquired through cameras and diagnostic tools that support maintenance inspections and decision-making processes. Natural language data comprise textual sources, including postflight reports, pilot complaints, technician notes and maintenance logs, providing valuable contextual information for fault identification and comprehensive system analysis.

The subsequent sections examine methods for efficiently storing and processing these heterogeneous data types in aviation PdM systems, with particular emphasis on scalable database technologies, cloud platforms, data-cleaning techniques to enhance analysis accuracy and reliability and AI algorithms tailored to each data modality to generate actionable insights.

4.3 | Different Databases

In aviation systems, data generally fall into three primary categories: internal data collected directly from onboard sensors and devices, such as temperature, pressure and velocity measurements; external data originating from sources outside the system, including weather and environmental conditions that influence performance and historical data derived from past system behaviour, providing insights into trends, patterns and potential vulnerabilities. Since DTs depend on continuous feedback between physical and virtual components, poor data accuracy or sensor failures can compromise real-time monitoring and may even result in physical system failures [41].

Within a DT framework, data exchanged between physical and virtual components are stored and managed using database systems tailored to internal, external and historical data. These data facilitate the creation of digital representation models as well as simulation and analysis tasks within the virtual system. The resulting simulations, recommendations and outputs are stored in databases and subsequently transmitted back to the physical system to enable appropriate actions or adjustments. Figure 5 illustrates the operations of DT technology, emphasising the interactions between physical and virtual systems through real-time and offline data managed across multiple databases. Collected and processed data may be structured, semistructured or unstructured and can be stored in various types of databases, including relational, nonrelational and multimodel databases.

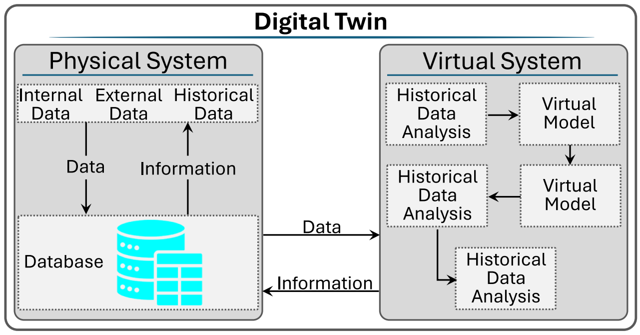

FIGURE 5 | General operations of a digital twin technology, how two physical and virtual systems interact with each other through databases which include different data types [41].

Relational databases organise structured data into tables representing entities (e.g., aircraft or sensor attributes), with rows corresponding to entity instances and columns storing attributes; each record is identified by a unique key. These databases utilise relationships between tuples to structure information and are accessed through structured query language (SQL), a standardised querying interface. Prominent relational database systems include MySQL, Microsoft SQL Server and Oracle [42, 43].

Nonrelational databases are specifically designed for semistructured data that does not fit into traditional tabular formats. These systems are optimised to handle massive data volumes by relaxing strict atomicity, consistency, isolation and durability (ACID) constraints, thereby prioritising performance and architectural flexibility over rigid accuracy. Prominent examples of NoSQL databases include MongoDB, Cassandra, CouchDB and HBase. MongoDB, for instance, enhances operational efficiency by supporting concurrent processing for rapid read/write tasks. Furthermore, it employs replica sets to ensure high availability and disaster recovery, which is essential for maintaining data integrity and supporting uninterrupted production operations [43, 44].

Managing diverse data types presents a significant challenge in modern data management systems. A multimodel database management system (MMDBMS) provides a unified platform for storing and managing multiple data models within a single system, thereby enabling efficient handling of such heterogeneous data [42, 45]. As represented in Table 2, this approach is particularly advantageous in aircraft systems, where data diversity is high and maintaining separate databases for each data type would be costly and complex. By enabling different data models to coexist within a single framework, a multimodel database facilitates integrated, faster and more efficient data access. The versatility of MMDBMSs, which support multiple data modelling paradigms, including key/value and graph-based models, renders them highly effective for DT technology applications [46].

TABLE 2 | Digital twin requirements for aircraft and assessment of database suitability [45].

DT requirement for aircraft	RDBMS	NoSQL	Multimodel databases (MMD)
Managing diverse and heterogeneous data	Low suitability	Medium suitability	High suitability
Handling large data volumes	Medium suitability	High suitability	High suitability
Real-time data processing	Medium suitability	High suitability	High suitability
Maintenance effort and schema evolution	Medium suitability	Low suitability	High suitability

These systems may rely on either a single database or multiple backend databases; however, when a single system lacks sufficient flexibility, heterogeneous databases may be required. By integrating and translating across various databases, MMDBMSs effectively address these limitations and provide a cohesive data management solution [47]. In large manufacturing environments, including aerospace, the massive volume of generated data makes full storage impractical, necessitating efficient data management techniques, such as automatic data aggregation to reduce storage requirements whilst supporting diverse data types, including documents, raw data and graphical models [48].

For example, ArangoDB is a schema-free multimodel database that supports document, key/value and graph data models, enabling the efficient storage of heterogeneous data in JSON format. Optimised for rapid storage of large datasets, ArangoDB is well-suited for large-scale and heterogeneous data projects. The master copy of data and models should be maintained in a centralised repository, serving as a reference for other instances. Offline instances may be employed for simulation and behaviour prediction under specific conditions, whereas online instances facilitate real-time remote monitoring of asset performance [49–51].

4.4 | Cloud Solutions for Aircraft Data in Digital Twins

Cloud computing plays a critical role in the deployment of DTs for PdM in aviation. This is primarily attributable to the immense volumes of diverse and complex data generated by aircraft systems, which necessitate scalable storage, high-performance processing and advanced analytical capabilities that cloud infrastructures inherently provide. These powerful cloud-based resources are essential for enabling the comprehensive modelling, learning, simulation and predictive functions that are critical to effective DTs, particularly in supporting proactive maintenance strategies [52]. This section discusses the significance of cloud computing solutions within the context of DTs for aircraft systems.

Through continuous updates with real-time or near-real-time data, DTs provide virtual replicas of physical aircraft systems or components to effectively monitor their condition. Cloud infrastructure is essential in this process as it enables large-scale storage and processing of sensor data from aircraft components, facilitating the creation of virtual models that deliver advanced analytical insights. By leveraging cloud computing, sensor data can be collected, stored and processed in near real-time, enabling PdM that proactively addresses potential issues, thereby enhancing safety and reducing operational costs. A recent study [53] illustrates a system in which cloud-based computing integrates IoT sensor data, aircraft lifecycle information and advanced algorithms into a DT ecosystem. Furthermore, cloud-based DTs enable the fusion of data from multiple sources, improving the accuracy and effectiveness of PdM models. Through cloud platforms, diverse sensor and system data can be aggregated and analysed collectively, providing a comprehensive view of the aircraft's condition. This integrated data processing is critical for developing robust predictive algorithms capable of forecasting component failures and optimising maintenance schedules. As highlighted by a recent review [54], the synergy between DTs and AI in a cloud environment supports the automation of PdM processes, leading to enhanced operational efficiency and reliability in aviation.

Real-time data processing is fundamental for PdM in aviation as it enables the early identification of potential issues and timely alerts to personnel, thereby improving decision-making. Additional benefits include enhanced safety, minimised downtime and optimised maintenance operations [54]. However, the real-time requirements of DTs pose a challenge as centralised cloud servers are often suboptimal due to the high volume of data exchanged between aircraft sensors and the server. In a study [55], the performance of DTs was assessed across three scenarios: cloud-only, fog–cloud hybrid and fog-only. The study concluded that fog computing significantly enhances response times, satisfying the real-time requirements of DTs. Furthermore, another study [56] proposed a three-layer architecture comprising edge, fog and cloud layers for the real-time and efficient processing of aircraft health monitoring algorithms. The edge layer is responsible for noise filtering and preprocessing, whereas the fog layer manages critical lightweight tasks, such as failure detection, using machine-learning algorithms. The cloud layer, by contrast, is designated for more complex and computationally intensive tasks, including data analysis, visualisation and the generation of feedback for the fog layer [57, 58]. Overall, these findings highlight that a hierarchical computing architecture, integrating edge, fog and cloud layers, provides an effective solution for meeting the real-time processing demands of predictive maintenance in aviation.

4.5 | Security Concerns and Solutions

One of the primary objectives of DT technology is to ensure seamless coordination between physical and digital domains as deviations in either domain can compromise the virtual representation and result in inaccurate predictions of the physical system [59, 60]. Because DTs establish a close relationship between physical assets and their virtual counterparts, they expose both physical and digital attack surfaces [19].

The physical attack surface encompasses threats to endpoints, such as cyber-physical systems (CPS), industrial internet of things (IIoT) devices and communication infrastructure, where vulnerabilities in the data acquisition and dissemination layers may be exploited to gain access to higher system layers. The collected data are highly sensitive as they contain critical operational and technical information about physical assets (e.g., aircraft), making them prime targets for both internal and external attackers. By contrast, the digital attack surface targets software, networks and computing resources, enabling malicious actors to infiltrate the DT, exfiltrate critical information and compromise essential data and system resources [19, 61].

The data management, synchronisation and data modelling layers are vulnerable to three primary categories of threats: attacks targeting computing infrastructure, attacks on virtual systems used for simulation and attacks against computational techniques for data management. Representative examples include man-in-the-middle (MitM) and data tampering attacks [62–64]. In MitM attacks, adversaries exploit network vulnerabilities during virtual machine or container migration or replication to intercept and manipulate virtual resources. In contrast, data tampering attacks involve unauthorised modifications throughout the data lifecycle, degrading data integrity, accuracy and granularity, and ultimately distorting the insights derived from the DT system. The data visualisation and accessibility layer relies on human–machine interfaces (HMIs) to present modelled data to end users, thereby supporting informed decision-making. The manipulation of HMIs represents a critical threat at this layer as unauthorised alterations to configurations can distort displayed information and result in erroneous interpretations by operators.

One promising approach to enhancing security in DTs involves leveraging cloud-based privacy mechanisms in conjunction with blockchain technology [61, 65]. Figure 6 illustrates a blockchain-based scheme designed to secure data integrity, prevent unauthorised modifications and ensure transparency in data access.

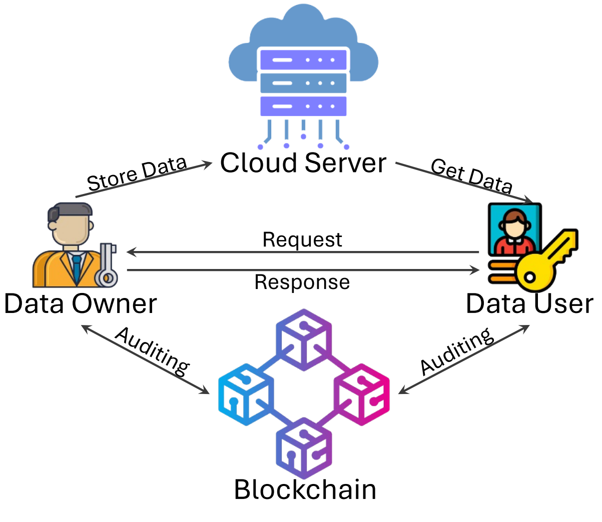

FIGURE 6 | Cloud-based privacy using blockchain scheme which can be used in digital twin technology [61].

One of the primary challenges in the secure sharing of DT data is ensuring data integrity verification, particularly in cloud computing environments where users may retrieve data from servers without an inherent mechanism for verifying integrity [66, 67]. Blockchain technology provides a robust solution for guaranteeing data integrity and verifiability by requiring validators to confirm the authenticity of newly submitted data before consensus is reached and the data are permanently appended to the ledger [68]. Once recorded, the data cannot be altered unless an attacker gains control of more than 51% of the validators—a scenario known as a 51% attack—which is generally considered practically infeasible in well-secured blockchain networks.

We present a brief description of a representative model from the literature, comprising four primary entities: the cloud server, the data owner/data collector, the data user and the blockchain [65]. The explanation and interpretation of this model are provided here for clarity and context.

In the aircraft PdM scenario:

- *Data Owner/Data Collector* is the airline company operates the aircraft and equips each with sensors (e.g., engine vibration, fuel flow, hydraulic pressure and temperature). It collects these sensor data in real time and securely transfers it to the cloud server after establishing an encryption key.
- *Data User* may be the aircraft manufacturer or a contracted maintenance provider requesting DT data to schedule repairs or order spare parts in advance. The data user sends a request to access the simulation results. The data owner verifies the request against blockchain records to ensure the user has permission.
- *Cloud Server* is a high-performance computing platform runs complex DT simulations of the aircraft. Using the live sensor data from the data owner, the cloud server creates and continuously updates the aircraft's DT. Simulation results (e.g., 'left engine bearing may fail in 20 flight hours') are sent back to the airline. If a maintenance company (data user) needs these data, the cloud server can share it, but only with the data owner's approval.
- *Blockchain* acts as a secure immutable ledger for all data transactions. Instead of storing the raw aircraft data itself, it stores hash values (digital fingerprints) of the data, along with access permissions and transaction logs. Every access request, approval or denial is recorded. Smart contracts automatically enforce access control rules.

In this scheme, the blockchain not only records access permissions but also governs data retrieval, with each transaction digitally signed through smart contracts to ensure verifiability and tamper resistance. Operating alongside the cloud server, the blockchain stores only the hash values of synchronised simulation datasets rather than the raw data, enabling detection of any data modification through hash verification. For example, when a DT simulation produces output D_{sim}, the airline (data owner) computes its hash and stores it on the blockchain as a data signature, whereas the complete simulation results are maintained on the cloud server.

When a maintenance provider (data user) submits a data request, the request is recorded on the blockchain and verified against predefined access control policies embedded in the smart contract. If authorised, the blockchain issues a one-time access key that allows the data user to retrieve D_{sim} from the cloud server; otherwise, unauthorised requests are detected, denied and permanently logged. Owing to the blockchain's immutable structure and consensus protocol, transaction records cannot be altered retroactively, ensuring long-term integrity and confidentiality of PdM data and associated access logs. For a comprehensive review of security issues in DTs, readers are referred to refs. [61, 69–71].

5 | Data Cleaning and Pre-Processing Techniques

Data play a central role in DT technology as applied to PdM. In a DT system, data pass through multiple stages, including data collection, preprocessing, storage and analysis. Data cleaning constitutes a critical component of this pipeline as real-world data are often incomplete, noisy, or inaccurate and must therefore be addressed during the preprocessing stage.

The selection of a cleaning technique is intrinsically linked to the underlying database architecture and data scale. For instance, traditional smoothing methods, such as SMA and Median filtering, are computationally efficient and well-suited for structured observations typically managed within relational databases (RDBMS) for smaller-scale diagnostic tasks. In contrast, highly dynamic and heterogeneous datasets—such as those generated by modern aero-engines and stored in multi-model or NoSQL databases—require more robust emerging techniques. These include Kalman filters for real-time state estimation and deep learning-based noise reduction, which are capable of handling the high velocity and variety characteristic of aviation big data within fog–cloud hybrid infrastructures. The practical utility of these cleaning techniques is evidenced by their integration into various aircraft health monitoring and digital-twin-related frameworks. In time-series analysis, data cleaning and normalisation are identified as critical preliminary stages for deep learning-based anomaly detection in complex aircraft systems, ensuring that sensor fluctuations do not lead to false alarms [72]. For graphical data, preprocessing steps, such as noise reduction and data augmentation, are essential for training robust CNNs used in automated visual inspections of aircraft structures [73]. Furthermore, in the context of natural language data, systematic cleaning—including the removal of linguistic noise and the standardisation of specialised terminology—is vital for the effective analysis of aviation safety reports and maintenance documentation, enabling the extraction of high-fidelity insights for safety management systems [74, 75].

In this section, we review the most common techniques for cleaning time-series, graphical and natural language data, which are essential for ensuring data quality in the application of DT technology to predictive maintenance PdM in aviation systems.

5.1 | Time-Series Data Cleaning Techniques

Time-series data in aviation, generated by sensors monitoring parameters, such as engine temperature, vibration and pressure, frequently contain noise—including outliers, sensor drift and random fluctuations—often exacerbated by environmental disturbances [2, 76]. Smoothing-based algorithms are essential for dampening these high-frequency fluctuations whilst preserving significant trends critical for fault diagnosis [77, 78]. As discussed in Section 2, the selection of these techniques is increasingly mapped to the data scale and the real-time requirements of the digital twin architecture.

Although the simple moving average (SMA) is effective for highlighting long-term trends in stable parameters, aviation-specific applications often prioritise the exponential moving average (EMA) and weighted moving average (WMA). These methods assign greater weight to recent data points, making them vital for monitoring dynamic systems, such as engine performance or velocity, where recent fluctuations are more indicative of impending failure than historical data [77].

To address impulsive noise and spikes caused by sensor malfunctions or environmental anomalies, such as turbulence, the median Filter is preferred. Unlike averaging methods, its robustness to extreme outliers allows for the removal of sharp spikes without blurring significant signal edges or degrading the overall power spectrum. For complex, safety-critical tasks, such as state estimation in aircraft navigation (e.g., tracking altitude or position), the Kalman Filter remains the industry standard. It provides optimal sensor fusion by recursively modelling system states and accounting for both process and measurement noise in real-time.

The choice between these techniques depends on the noise characteristics: moving averages (SMA, EMA and WMA) excel at reducing Gaussian noise, whereas median filters are superior for impulsive spikes. These strategies are further summarised in Table 3, which maps each algorithm to its specific advantages and aviation use cases.

5.2 | Graphical Data Cleaning Techniques

Raw image and video data frequently contain noise, missing values, inconsistent formats or irrelevant information, all of which can significantly impair the performance of machine learning algorithms. Consequently, data cleaning and preprocessing are essential steps to ensure high-quality data for subsequent processing and analysis. These procedures offer multiple benefits for image and video data. Addressing quality issues helps mitigate the risks of model overfitting or underfitting. Moreover, techniques, such as data augmentation, can effectively increase dataset size and diversity, which is particularly advantageous when working with limited data. Efficient preprocessing also accelerates training as well-structured data can be processed more effectively by machine learning models. In the following, we outline some of the most commonly

TABLE 3 | Summary of common techniques used for noise reduction in time-series data [77].

Algorithm	Typical usage	Advantages	Limitations	Additional considerations
SMA	Basic smoothing of noisy sensor signals over short windows	Easy to compute; effective for reducing short-term fluctuations	May blur sharp features (e.g., peaks or abrupt changes); fixed window limits adaptability	Computationally efficient; suitable when long-term trends are of primary interest
EMA	Smoothing time-series data with gradual trends (e.g., temperature or sensor drift)	Simple to implement; assigns higher weight to recent samples; adapts to trends	May lag during rapid changes; sensitive to smoothing factor selection	Memory efficient; performance strongly depends on the choice of the smoothing parameter (α)
WMA	Applications where recent observations are more informative than older ones	Flexible weighting scheme; more adaptive than SMA	Requires domain knowledge for weight design; higher computational cost than SMA	Well suited for application-specific sensor characteristics
Median filtering	Removal of impulsive noise and outliers in sensor measurements	Effectively suppresses spikes; preserves edges better than averaging methods	May over-smooth clean signals; performance depends on window size	Nonlinear method; particularly useful for faulty or unstable sensors
Kalman filter	Real-time state estimation and sensor fusion in dynamic systems	Optimal for linear Gaussian systems; supports real-time estimation and fusion	Implementation complexity; requires system model and noise statistics	Well suited for time-varying systems; computationally demanding for large-scale deployment

employed data cleaning and preprocessing techniques for image and video data.

- *Noise Reduction*: Noise is a major issue affecting image and video data, often caused by low-light conditions, compression artefacts or sensor limitations. To mitigate or eliminate noise, various techniques, such as Gaussian filtering, median filtering and Wavelet transformations, are applied to enhance data quality. These methods not only smooth the data but also preserve essential characteristics such as edges and textures [79].
- *Normalisation and Standardisation*: This process involves rescaling pixel values of images or video frames to a consistent range, such as [0,1] or $[-1, 1]$. Normalisation ensures uniformity across datasets, allowing models to train and generalise more effectively. Standardisation, on the other hand, adjusts pixel values to have a mean of zero and a standard deviation of one, facilitating faster and more stable model convergence.
- *Removing Duplicate or Irrelevant Frames*: Video datasets often contain redundant or irrelevant frames, increasing computational costs and reducing processing efficiency. Deep learning techniques can identify and retain only the most informative frames, ensuring that the dataset focuses on key moments where significant changes or events occur [80].
- *Annotation Verification and Correction*: In supervised learning tasks, inaccurate or inconsistent annotations can adversely impact model performance. Ensuring precise annotations is essential for graphical data processing. Automated annotation verification and correction frameworks streamline this process, improving the reliability of both large-scale and smaller datasets [81].
- *Cropping and Padding*: Cropping removes irrelevant or noisy regions from images to focus on areas of interest, whereas padding standardises image dimensions to meet model input requirements. These techniques are particularly important in object detection tasks, where precise cropping of target objects enhances model accuracy and consistency.
- *Data Augmentation*: Data augmentation is a critical technique for enhancing image and video datasets, particularly in fields, such as aviation, where machine-learning models support applications such as aircraft auto-landing systems. By artificially increasing dataset diversity through methods, such as image transformation and synthesis, models can better generalise to different objects and scenarios, improving safety and reliability [82].

5.3 | Natural Language Data Cleaning Techniques

Unstructured data sources, including maintenance documentation, technical manuals, pilot reports and inspection notes, contain valuable information relevant to aircraft operations, fault logs and repair activities. These data are critical for the efficient functioning of MRO departments within the aviation industry. Proper analysis of MRO data can significantly enhance aircraft safety and performance by enabling improved tracking of defect recurrence and supporting effective component lifecycle management through PdM.

NLP techniques enable the extraction of actionable insights from large volumes of unstructured MRO data. Text mining and entity recognition support PdM strategies by identifying early indicators of mechanical issues. For example, NLP applications that leverage text data from maintenance and safety reports enhance aviation safety management systems [16]. Processing natural language data from maintenance records and reports is essential for informed decision-making and operational efficiency within the aviation MRO sector. Key data cleaning techniques tailored to this domain include:

- *Standardising Domain-Specific Terminology and Abbreviations*: Aviation maintenance records frequently contain specialised terms and abbreviations. Developing comprehensive dictionaries or ontologies to standardise these terms is crucial. For example, a domain-specific ontology ensures accurate interpretation of maintenance records [83].
- *Noise Removal*: Maintenance logs often include irrelevant information, such as nonstandard spelling, grammatical errors or extraneous symbols. Implementing preprocessing steps to filter out this noise enhances data quality. For instance, NLP tools can be employed to clean maintenance records, improving their clarity and usability [84].
- *Entity Recognition and Standardisation*: Identifying and standardising key entities, such as component names or fault types, is vital for efficient data utilisation. Named entity recognition (NER) techniques tailored to aviation maintenance can extract meaningful information from textual records. For example, applying NER to maintenance logs aids in structuring and analysing data effectively.
- *Data Structuring*: Converting unstructured text into structured formats enables more efficient analysis and integration with analytical tools. This process involves parsing narratives into predefined fields or categories. For instance, structuring maintenance data into a standardised format facilitates enhanced data mining and predictive analysis [39].

Implementing these data cleaning techniques ensures that natural language data in aviation MRO is accurate, consistent and well-structured, ultimately supporting PdM strategies and operational improvements.

6 | AI Algorithms for Different Types of Data in Predictive Maintenance

Maintenance costs constitute a significant challenge within the aviation industry. According to the international air transport association (IATA) [85], global MRO expenditures reached $78.6 billion in 2022, accounting for approximately 11% of total airline operating costs. Owing to its ability to process large volumes of historical data and generate predictive insights, AI offers a promising approach to mitigating these costs. Specifically, AI enables maintenance teams to proactively schedule preventive interventions before issues escalate into critical failures, thereby reducing aircraft downtime and the costs associated with component malfunctions. Consequently, the integration of AI into maintenance operations is increasingly recognised as a key strategy for enhancing the long-term sustainability of the aviation sector [86–88].

PdM in the aviation sector primarily adopts two methodological paradigms: model-based and data-driven approaches. Model-based methods rely on mathematical and physical representations to simulate the behaviour of individual components; however, the complex interdependencies among aircraft subsystems often constrain their predictive accuracy, limit adaptability and necessitate substantial manual calibration and expert intervention [79, 89]. In contrast, data-driven approaches leverage machine learning and deep learning techniques to effectively capture the complexity and dynamic behaviour of aircraft systems by operating directly on observed data, without requiring detailed prior knowledge of underlying mechanical processes. Moreover, the widespread deployment of IoT technologies, together with the increasing availability of large-scale operational and maintenance datasets, has further strengthened the applicability of data-driven models, enabling more accurate modelling of system degradation and robust assessment of component health in real-world aviation environments [90, 91].

In the following, we systematically review AI and deep learning methodologies applied to time-series data (for RUL and fault detection), graphical data (for fuselage and borescope inspection) and natural language data (for safety and maintenance reports).

6.1 | AI Algorithms for Time-Series Data

Recent advancements in aviation technology have enabled the continuous monitoring of aircraft components via sensors that generate time-series data capturing key operational parameters, such as temperature, pressure and velocity. These data are fundamental for assessing the operational state of aircraft systems and play a critical role in tasks such as pattern recognition, anomaly detection and prognostics. When applied to such data, AI algorithms enhance the safety and reliability of critical components, including engines and actuators, thereby reducing the likelihood of catastrophic failures and unplanned flight disruptions. The analysis of aviation time-series data generally concentrates on two primary objectives: (1) RUL estimation, which aims to predict the time remaining before a component is expected to fail, and (2) fault detection, which seeks to identify incipient malfunctions at an early stage to ensure operational safety and system reliability.

6.1.1 | Remaining Useful Life Estimation

As a fundamental element of aviation prognostics and health management (PHM), RUL estimation facilitates proactive maintenance strategies aimed at reducing costs and operational risks [92]. Data-driven approaches have become the dominant paradigm for this task [93], with recurrent architectures—specifically long short-term memory (LSTM) networks—and shallow CNNs being the primary tools for processing multivariate flight data [8, 94, 95]. Hybrid models combining LSTM with attention mechanisms have shown superior predictive performance by prioritising informative temporal segments in engine sensor streams [8]. Furthermore, integrating these RUL estimates into maintenance scheduling has directly optimised decision-making in fleet management [94]. To address the limitations of uniform update mechanisms in standard LSTMs, the multicellular LSTM (MCLSTM) architecture was developed for aero-engines, utilising five hierarchical levels to capture both long-term degradation and short-term operational variations [96]. Similarly, hybrid CNN-transformer frameworks have been introduced to model inter-sensor dependencies; for instance, the adaptive graph convolutional transformer encoder (AGCTE) utilises GCNNs to process graph-structured sensor data while an encoder-only transformer models long-term temporal dependencies [97, 98]. Recent comparative studies on aviation engines indicate that classification-based models often outperform regression approaches, offering lower RMSE and AUC with reduced computational overhead [99–101]. Advanced ensemble methodologies integrating XGBoost, GRU and LSTM with voting-based feature selection have been employed to capture complex spatial and temporal characteristics [101]. To improve model generalisation, techniques, such as random sampling-based class balancing (RSBCB), have been successfully applied to address the inherent scarcity of failure data in aero-engine datasets [101]. A significant portion of RUL research focuses on aero-engines due to their safety-critical nature and high degradation rates under extreme operating conditions [95]. The NASA-developed C-MAPSS dataset serves as the primary benchmark, providing multivariate time-series from 21 sensors across four distinct fault scenarios (FD001–FD004) to facilitate model training and cross-study evaluation [102]. Model performance is typically quantified using RMSE and the scoring function (Score); the latter is specifically valued in aviation as it penalises late predictions that could lead to unscheduled downtime or safety incidents [102]. Table 4 summarises the specific model performances reviewed in this context.

6.1.2 | Fault Detection

Fault detection is essential for identifying incipient failures in aircraft engines, bleed air systems and bearing assemblies by extracting informative features from raw time-series data. Existing methodologies are primarily categorised into reconstruction-based, prediction-based and classification-based approaches.

Reconstruction-based methodologies detect faults by monitoring deviations from learnt normal behaviour [103, 104]. This approach has been implemented using LSTM networks and autoencoders (AE) for air bleed systems, where substantially higher reconstruction errors signal abnormal sequences [103]. Similarly, GCNNs integrated with LSTM autoencoders have achieved 97.3% accuracy in predicting bearing faults on the CWRU dataset, demonstrating applicability to aviation systems after suitable adaptation [104, 105].

Prediction-based approaches identify potential faults through significant deviations between forecasted and observed system states [9, 106]. In this context, informer models have demonstrated superior performance over standard LSTM networks for long-sequence time-series forecasting [9]. Additionally, multimodal regularised LSTM architectures processing both raw and SSA-smoothed data have enabled the detection of piston-engine anomalies, including flameouts, up to 2 min prior to occurrence [106, 107].

Classification-based approaches utilise supervised learning, with algorithms, such as support vector machines (SVM) and random forests, widely adopted for high-dimensional aviation problems. The isolation forest algorithm [108] is frequently applied to aero-engines due to its ability to detect anomalies without labelled fault data and its suitability for airborne computing platforms [91, 109]. Enhancements involving wavelet-based denoising and weighted feature selection have

TABLE 4 | Summary of data-driven models proposed for remaining useful life (RUL) estimation of aero-engine components using the C-MAPSS dataset.

Ref. (year)	ML/DL approach	Main contribution	RMSE	Score
[8] (2020)	LSTM with attention	Attention-based temporal feature weighting	20.81	5971.58
[96] (2021)	MC-LSTM with attention	Trend-aware state update mechanism	18.76	2570.50
[97] (2022)	CNN with transformer	Improved modelling of long-term temporal dependencies	15.65	951.00
[94] (2024)	CNN–LSTM with attention	Bayesian optimisation for hyperparameter tuning	14.19	1056.21
[98] (2024)	Graph CNN with encoder-only transformer	Explicit modelling of intersensor relationships	13.74	746.32
[99] (2024)	Multiple ML models	Comparative evaluation of classical ML approaches	10.90	N/A
[100] (2025)	CNN–Transformer–LSTM	Reinforcement learning-based hyperparameter tuning	11.95	634.36
[101] (2025)	Multiple ML/DL models	RSBCB and voting-based feature selection	12.61	N/A
[95] (2025)	CNN with Bi-LSTM and attention	Integration of RUL prediction into maintenance planning	13.96	3433.06

Note: The dataset consists of four subsets (FD001–FD004), with most studies reporting results on either two subsets (commonly FD001 and FD004) or all four. The table reports the average performance metrics as presented in the original studies; unreported metrics are indicated as N/A.

further improved its effectiveness in detecting turbofan engine failures [109, 110].

SVM-based techniques remain prevalent in academic research due to their accuracy on small datasets [111, 112]. Hybrid architectures combining LSTM with one-class SVMs have been proposed to identify abnormal mechanical behaviour from vibration measurements using physical understandings of fault-induced signals [113]. Recent research also employs two-tiered LSTM frameworks to mitigate misclassification in overlapping fault characteristics by specifying fault types across sequential flight data like velocity and pitch rate [114]. Furthermore, image-based detection for aerospace sensors has been improved through pretrained vision models, such as VGG16, combined with data augmentation strategies [115].

Aviation fault detection remains challenged by system complexity, noisy environment and the severe imbalance of real fault data [91]. Mitigation strategies involve combining real flight records with simulated data and benchmark datasets such as CWRU [105, 116], Paderborn [117], MFPT [118] and NASA's C-MAPSS [102]. However, security and confidentiality concerns continue to limit the availability of publicly accessible aviation datasets [104]. Table 5 summarises the developed models for various aircraft components.

6.2 | AI Algorithms for Graphical Data

Graphical data, including images, videos, 3D models, thermograms and spectrograms, plays an increasingly significant role in aviation, particularly in visual inspection and MRO operations. Such data are acquired through a range of methods, from human-assisted observation to advanced sensing technologies, including UAS, infrared thermography, LiDAR and borescopes, and supports the detection and evaluation of structural and surface defects in critical aircraft components such as the fuselage, wings, skin and engine blades [14]. Traditionally, visual inspections have been conducted manually by trained personnel, a process that is subjective, labour-intensive and time-consuming, and is affected by human factors such as fatigue, environmental conditions and limited visibility. These limitations can result in missed or misclassified defects, particularly those smaller than 1 mm [119]. Moreover, as aircraft systems increase in size and complexity, the scalability constraints of manual inspections become more pronounced, emphasising the need for automated inspection solutions to enhance reliability, reduce costs and support sustainable aviation maintenance practices [15, 81].

Automated visual inspections for aircraft are generally categorised into two types: (1) fuselage inspection and (2) borescope inspection. Fuselage inspection, often focusing on the exterior structure, aims to identify surface defects such as paint detachment, cracks, corrosion and damages from impacts [120]. In contrast, borescope inspection is a specialised visual method used primarily for internal components, such as aero-engine turbine blades, allowing technicians to inspect hard-to-access areas without disassembling the engine [121]. For illustrative purposes, Figures 7 and 8 depict sample images obtained from fuselage and borescope inspections, respectively.

TABLE 5 | Summary of data-driven fault detection models for time-series data.

Ref. (year)	Component	Main contribution	ML/DL approach	Accuracy
[103] (2021)	Air bleed system	Unsupervised representation learning for anomaly detection	LSTM autoencoder	N/A
[91] (2021)	Aero-engine	Fault detection without requiring large volumes of faulty data	Isolation forest	N/A
[113] (2022)	Gears and bearings	Semisupervised anomaly detection framework	LSTM with SVM	0.88
[115] (2023)	Sensors	Integration of explainability analysis into fault detection	CNN	0.99
[106] (2024)	Piston engine	Regularisation-based improvement for fault detection	LSTM	N/A
[9] (2024)	Precooler and pressure regulating valve	Temporal modelling using an informer architecture	Informer	0.96/0.93
[104] (2024)	Bearing	Simulation-based fault data generation for training	GCNN with LSTM autoencoder and self-attention	0.97
[109] (2024)	Aero-engine	Extreme-difference-based feature selection strategy	Isolation forest	N/A
[112] (2024)	Actuator	Demonstration of SVM effectiveness for fault detection	SVM	0.92
[114] (2025)	Actuators and propulsion system	Two-tier fault diagnosis architecture	LSTM	0.99

Note: All evaluations were conducted on custom datasets, with no common benchmark shared across studies. Reported performance metrics are taken directly from the original sources; unreported values are indicated as N/A.

FIGURE 7 | Sample images obtained from fuselage inspection [120]. These images were collected either manually or by drones from various parts of A320 and B737 aircraft undergoing maintenance, including the fuselage, wings and tail. The defects depicted in these samples fall into two main categories: surface paint detachment and surface scratches.

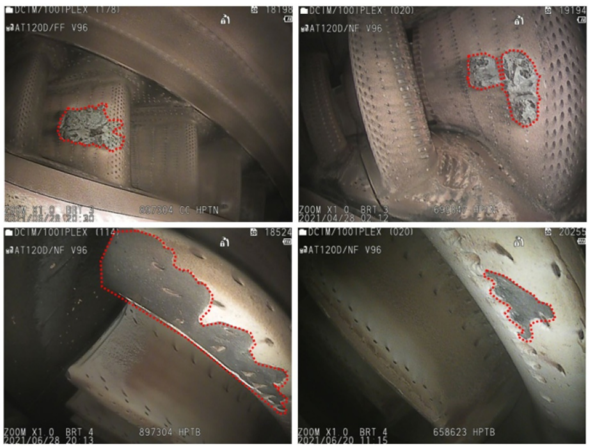

FIGURE 8 | Sample images obtained from borescope inspections [122]. These images presents the detection results of the YOLOv7 model for the same types of defects but with varying shapes and appearances. The four sub-images, arranged from left to right and top to bottom, show: top-left and top-right: examples of corrosion (with different patterns and severities); bottom-left and bottom-right: examples of Thermal Barrier Coating (TBC) missing (in various forms and locations on the blade surface).

6.2.1 | Visual and Structural Inspection

Visual inspection, encompassing both external fuselage surfaces and internal engine components via borescope, is essential for identifying structural defects such as cracks, corrosion and deformations. Recent methodologies have integrated mask region-based convolutional neural networks (Mask R-CNNs) [123] for simultaneous object detection and instance-level segmentation of fuselage defects [81]. Strategies, such as dataset balancing, augmentation and the use of SVM-based preclassifiers, have significantly improved prediction accuracy for nondefective images [81]. For corrosion detection, DenseNet architectures have demonstrated high accuracy when processing data from the d-Sight aircraft inspection system [124] as illustrated in

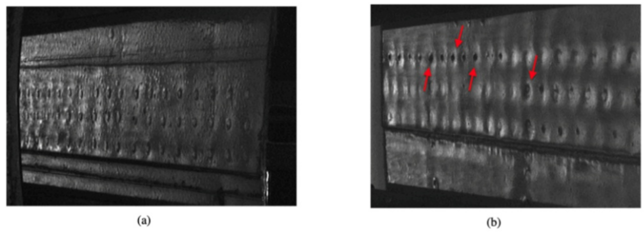

FIGURE 9 | Two samples taken with the DAIS 250c device of a simple shear lap joint. (a) An image with no corrosion and (b) the corrosion pillowing affecting rivets in red colour [124].

Figure 9. Furthermore, the integration of visual attention modules into architectures, such as the attention residual-U-Net, has enhanced feature localisation along weak boundaries [120, 125, 126]. Drone-based systems have also been introduced to automate high-resolution data acquisition, facilitating inspection in difficult-to-reach areas within Aviation 4.0 frameworks [127, 128].

For internal engine assessment, AI-based image analysis facilitates the automation of borescope inspections [129]. Coarse-to-fine frameworks utilise deep CNNs to filter background regions before precise defect localisation, optimising computational efficiency [119]. Specialised modifications to Mask R-CNN, incorporating multiscale feature fusion and balanced loss functions, have further improved damage detection in turbine blades [130]. Real-time inspection capabilities are predominantly supported by YOLO-based frameworks (v5 and v7), which employ deformable convolutions and Haar wavelet-based augmentation to maintain high-frequency details in complex backgrounds [122, 131, 132]. Additionally, hybrid deblurring strategies using Pix2Pix GANs have been proposed to mitigate the poor image quality often encountered in borescope feeds [133]. Table 6 provides a summary of the models developed for fuselage and borescope inspections.

6.3 | AI Algorithms for Natural Language Data

Textual data from sources, such as incident reports, maintenance logs and pilot feedback, provide critical contextual information for flight safety. However, manual analysis is highly resource-intensive; for instance, reviewing a single incident report can require up to five working days [134]. Given that human factors contribute to a significant proportion of accidents, scalable analytical methods are necessary to identify safety-critical patterns such as fatigue or stress [135]. AI and NLP techniques address this challenge by automating the extraction of actionable insights for proactive risk mitigation.

Aviation safety research primarily utilises the aviation safety reporting system (ASRS) [136] and the national transportation safety board (NTSB) [137]. ASRS is a proactive voluntary reporting programme focused on systemic risks and human factors, whereas NTSB reports stem from reactive investigations into accident causes and recommendations. Despite their differing reporting mechanisms, both datasets are foundational for systematic safety research and large-scale textual analysis [16, 75].

TABLE 6 | Summary of data-driven models proposed for graphical data analysis in aircraft inspection tasks.

Ref. (year)	Inspection task	Main contribution	CNN arch.	Acc.	Prec.	Rec.	F1
[81] (2020)	Fuselage inspection	Image preclassification for false-positive reduction	Mask R-CNN	N/A	0.713	0.641	0.675
[124] (2021)	Fuselage inspection	DAIS-generated synthetic images for training enhancement	DenseNet	0.922	0.932	0.933	0.914
[119] (2021)	Borescope inspection	Coarse-to-fine defect detection strategy	N/A	0.935	0.948	0.961	0.954
[130] (2022)	Borescope inspection	Feature enhancement using TFNet and BL modules	Mask R-CNN	Scenario-specific	N/A	N/A	Scenario-specific
[122] (2022)	Borescope inspection	Integration of DCN and DSC modules into the detection pipeline	YOLOv5s	N/A	0.933	0.762	N/A
[120] (2022)	Fuselage inspection	Incorporation of attention mechanisms	Mask R-CNN	N/A	N/A	N/A	N/A
[133] (2023)	Borescope inspection	GAN-based motion deblurring and image enhancement	U-Net	0.998	0.931	0.955	N/A
[125] (2024)	Fuselage inspection	Residual blocks combined with attention mechanisms	U-Net	0.938	N/A	N/A	N/A
[127] (2024)	Fuselage inspection	Multiclass defect detection across five defect categories	YOLOv8s	0.835	N/A	N/A	N/A
[132] (2025)	Borescope inspection	Integration of HWFA and channel-attention blocks	YOLOv7	N/A	Per-class	Per-class	N/A

Note: Fuselage inspection studies focus on the aircraft fuselage, skin, wings and tail sections, whereas borescope inspection studies target aero-engine blades, vanes and compressor components. Performance metrics are reported as presented in the original studies; scenario-specific or per-class results are indicated accordingly, and unreported metrics are denoted as N/A.

AI solutions in this domain are categorised into DL and LLM based approaches. Various DL architectures, such as LSTM and CNN, have been extensively adopted to classify flight phases and identify risk levels from unstructured narratives [138, 139]. More recently, LLMs integrated with retrieval-augmented generation (RAG) frameworks have enabled the synthesis of information from diverse sources to provide contextually relevant support for pilot decision-making and aviation training [140, 141].

6.3.1 | Deep Learning Based Models

Predictive models have been developed to forecast the severity and damage potential of adverse events using historical narratives and event sequences from the NTSB dataset [142]. For instance, a model employing word embeddings and LSTM networks achieved accuracies of 80% and 77% for the binary classification of accidents versus incidents [142]. To enhance risk management, ensemble architectures have also been proposed to automatically determine the risk level of unsafe events [139]. This approach integrates a CNN for structured data feature extraction with a bidirectional LSTM enhanced by an attention mechanism (Att-BiLSTM) for unstructured narratives. Through a model fusion strategy, this ensemble achieves an 87% accuracy and demonstrates high robustness when processing imbalanced aviation datasets [139].

6.3.2 | LLM Based Models

LLMs, primarily based on the transformer architecture, are developed through a two-phase process: pretraining on large-scale corpora followed by fine-tuning on domain-specific tasks. Owing to their scalability and generalisation capabilities, these models have become foundational tools for safety-critical sectors, including aviation [143, 144].

A study proposed the ASRS–CMFS model, a compressed adaptation of the RoBERTa architecture [145]. Pretrained on ASRS data and fine-tuned on 14 specific classification tasks, ASRS–CMFS contains significantly fewer parameters (one-seventh) and lower computational costs compared to RoBERTa—reducing FLOPS by a factor of 489 during pretraining and 5 during inference—making it highly efficient for real-world deployment. Similarly, SafeAeroBERT [146] utilises the BERT architecture pretrained on ASRS and NTSB data. By focusing on aerospace-specific language, it outperformed both BERT and SciBERT in identifying contributing factors within incident reports.

Research has also examined general-purpose LLMs for aviation safety and human factors analysis [147, 148]. For instance, GPT-3.5 has been evaluated for automating ASRS tasks, such as synopsis generation and responsibility assessment, producing summaries comparable to human outputs despite limitations in complex reasoning [147]. More advanced models, such as GPT-

4o, when integrated with frameworks, such as the Human Factors Analysis and Classification System and Chain-of-Thought reasoning, have achieved human-expert performance in certain inference tasks involving unsafe acts [148]. These findings suggest that although not yet suitable for direct deployment, LLMs serve as powerful decision-support tools.

RAG frameworks [149] mitigate LLM limitations, such as hallucinations and lack of domain knowledge, by grounding generative models in external knowledge bases such as maintenance manuals and regulatory documents [140, 150, 151]. For instance, a RAG-based workflow using ATA100 and aircraft maintenance manuals improved classification accuracy for defective components by retrieving relevant technical data before generating summaries [140]. Additionally, integrating LLMs with the X-Plane flight simulator has enabled context-aware real-time assistance during aviation emergencies [150].

The integration of LLMs with deep learning techniques further enhances information extraction from fragmented aviation data. For example, an Intelligent Aviation Safety Hazard Identification Method utilises a hybrid BERT, Bi-LSTM and CRF model to identify entities from incident texts, achieving an F1 score of 94.18% [152]. Such models, combined with knowledge graphs, improve interpretability and support proactive safety management.

Despite their potential, applying NLP in aviation maintenance is challenged by domain-specific terminology, data scarcity and the need for high explainability. Future research is expected to focus on aviation-specific LLMs, RAG integration with knowledge graphs and human-in-the-loop systems to ensure trustworthy decision-making [153, 154]. Table 7 summarises the models developed for natural language tasks within this review.

7 | Challenges and Future Research Directions

Although the integration of DT and AI offers transformative potential for aviation PdM, several critical barriers must be addressed to transition from research to operational deployment. These challenges are organised into three primary themes: data sovereignty and cybersecurity, real-time synchronisation and AI interpretability.

7.1 | Data Sovereignty and Cybersecurity in Shared Aviation Ecosystems

Aviation PdM operates within a multistakeholder ecosystem involving original equipment manufacturers, airlines and MRO providers. A fundamental challenge is *data sovereignty*: the tension between the necessity of data sharing to improve AI accuracy and the protection of proprietary engineering data. The sensitive nature of these datasets requires robust encryption and privacy-preserving mechanisms throughout the data lifecycle [19, 155]. Compliance with regulations, such as the GDPR and the protection of sensitive flight, records demand advanced anonymisation strategies and continuous investment in cybersecurity frameworks [66, 156].

To mitigate these concerns, future research should explore the implementation of federated learning (FL), which allows models to be trained across multiple decentralised servers

TABLE 7 | Summary of data-driven models proposed for natural language data in aviation safety applications.

Ref. (year)	Dataset	Main contribution	DL approach/LLM	Application aim	F1
[142] (2021)	NTSB	Temporal modelling of event sequences in safety reports	Word embeddings with LSTM	Prognosis of adverse events	0.780
[145] (2022)	ASRS	Pretraining a compact RoBERTa model on domain-specific reports	RoBERTa	Report classification	0.535
[139] (2022)	ASRS	Model fusion for robust aircraft risk identification	CNN with Att-BiLSTM	Aircraft risk identification	0.840
[146] (2023)	ASRS and NTSB	Domain-specific pretraining of BERT on aviation reports	BERT	Report classification	0.656
[147] (2023)	ASRS	Exploration of large generative models for safety analysis	GPT-3.5	Aviation safety analysis	N/A
[152] (2024)	Maintenance risk reports (China)	Graph-based safety modelling using Neo4j	BERT–LSTM–CRF	Safety hazard identification	0.942
[140] (2024)	N/A	Retrieval-augmented generation for dynamic knowledge integration	Qwen 1.5	Accident investigation	N/A
[150] (2025)	N/A	Retrieval-augmented generation for decision support	GPT-4o	Pilot assistance	N/A
[148] (2025)	NTSB	Chain-of-thought reasoning for safety analysis	GPT-4o	Aviation safety analysis	0.513
[151] (2025)	NTSB	Retrieval-augmented semantic representation learning	Sentence-BERT and Qwen 2	Aviation safety analysis	N/A

Note: Performance metrics are reported as presented in the original studies; unreported values are indicated as N/A.

without exchanging raw data. Additionally, the integration of Blockchain technology can provide a decentralised immutable ledger for maintenance history, ensuring data integrity and transparent access control without compromising proprietary interests.

7.2 | Real-Time Synchronisation Between Physical Assets and Digital Twins

A core requirement of a functional DT is maintaining high-fidelity synchronisation with its physical counterpart. However, aircraft systems generate massive volumes of high-velocity sensor data [25, 157]. The 'fidelity gap' occurs when communication latencies or computational bottlenecks prevent the digital model from updating at a rate that faithfully reflects the current state of the in-flight asset [56, 158]. Achieving near-zero latency requires an orchestrated edge–fog–cloud infrastructure [55].

Future efforts should focus on adaptive sampling and compression algorithms that prioritise safety-critical data packets during low-bandwidth phases. Furthermore, the development of on-device model compression (e.g., pruning and quantisation) will enable more complex DT simulations to run directly on airborne edge hardware, minimising the need for constant cloud synchronisation and reducing the impact of transmission latencies.

7.3 | Interpretability of AI Models for Safety-Critical Decision Making

The 'black-box' nature of deep learning architectures remains a significant barrier to regulatory certification and operational trust. In safety-critical aviation tasks, maintenance personnel must understand the rationale behind an AI-generated RUL estimate or fault alarm [103, 159]. Domain shifts—resulting from differences in engine types or operational changes—can further degrade the reliability of these nontransparent models [160].

The next generation of aviation AI must move towards explainable AI (XAI) frameworks that provide visual or natural language justifications for maintenance alerts. Moreover, the integration of physics-informed neural networks represents a promising path forward; by embedding physical laws and engineering constraints directly into the AI architecture, models become more interpretable, easier to validate for certification and more robust when handling out-of-distribution operational data.

8 | Conclusions

PdM represents a fundamental shift in aviation maintenance strategies by enabling proactive condition-based decision-making that enhances safety, reliability and operational efficiency. Unlike traditional schedule-based or reactive approaches, data-driven PdM leverages heterogeneous data streams—ranging from time-series sensor measurements and inspection imagery to unstructured textual reports—to anticipate failures and optimise maintenance actions across the aircraft lifecycle.

This study provided an integrated and aviation-specific review of PdM by jointly examining engineering data management, DT architectures and AI algorithms as interdependent components of a unified maintenance pipeline. By synthesising these dimensions, the study highlighted how data quality, preprocessing, storage architectures and model selection directly influence DT fidelity and the reliability of AI-based prognostic and diagnostic outcomes. In contrast to prior reviews that focus on isolated technologies, this work emphasised cross-layer dependencies and system-level considerations that are critical for real-world deployment in safety-critical aviation environments. Although advanced AI and deep learning models have achieved promising results in RUL estimation, fault detection, visual inspection and text-based safety analysis, their effectiveness is constrained by persistent challenges. These include limited access to real-world failure data, data imbalance, scalability and latency issues in real-time settings, cybersecurity and privacy concerns and the lack of explainability in complex models. Addressing these challenges requires not only algorithmic advances but also robust engineering data management solutions, such as multimodel databases, fog–cloud hybrid infrastructures and privacy-enhancing technologies. Looking forward, future research should prioritise the development of explainable and certifiable AI models, standardised data frameworks for aviation PdM and high-fidelity DTs that seamlessly integrate physics-based models with data-driven learning. Greater collaboration between academia, industry and regulatory bodies will be essential to bridge the gap between experimental studies and operational deployment. By providing a structured end-to-end perspective, this paper aims to serve as a foundation for advancing safer, more efficient and scalable predictive maintenance systems in the aviation industry.

Author Contributions

Saber Mehdipour: conceptualisation, resources, writing – review and editing, writing – original draft. **Ali Habibzadeh:** conceptualisation, writing – original draft, writing – review and editing, resources. **Nima Esmi:** methodology, writing – review and editing, visualisation, project administration, resources. **Paria Heravi:** writing – original draft. **Mahdiyeh Mirsaraee:** writing – original draft. **Asadollah Shahbahrami:** writing – review and editing, conceptualisation. **Zeinab Javanmardi:** conceptualisation, writing – review and editing.

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